

Progress Report: Sequence AI Agent

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Completed Work

Phase 1: We built a complete, runnable Sequence game engine including board layout, deck management, move validation for all card types (regular, two-eyed Jack wild, one-eyed Jack removal), win detection with corner wilds, and a game loop with a random baseline agent. This foundation is validated through hundreds of automated games.

Phase 2: We implemented a heuristic evaluation function that scores board states using four weighted features, sequence progress with exponential scaling ; 4-in-a-row scores 125 and a 3-in-a-row scores 25, opponent threat assessment using the same analysis applied to the opponent's chips, board control based on how many potential sequences pass through each occupied cell, and Jack utility that recognizes when one-eyed Jacks become critical for blocking near-complete opponent sequences. This heuristic drives a minimax search engine with alpha-beta pruning, move ordering that prioritizes high-value positions to maximize pruning efficiency, and a configurable move cap to keep decision times practical.

Phase 3: We built the probabilistic reasoning layer that handles the hidden information problem. A card counting module tracks the unknown pool by subtracting the agent's hand and all discarded cards from the full double deck. The sampling function draws plausible opponent hands with probability proportional to remaining card copies, following the belief-guided approach from Yao et al [2]. The determinization function converts each sample into a fully observable state for minimax search, and the policy averaging pipeline inspired by AlphaZero's [1] approach. We combine search results across multiple samples, so no single determinization dominates the final decision.

Phase 4: We connected all components into a single combined agent. At each decision point, the agent generates 3–5 determinized states from the belief model, runs minimax on each, averages the move scores, and plays the highest-scoring move. The integrated agent consistently beats the random baseline in tournament play, confirming the pipeline works end-to-end.

To do:

Phase 5 (Self-Play RL + Greedy Baseline): The heuristic currently uses hand-tuned weights. We will optimize them through self-play reinforcement learning, the agent plays against copies of itself with perturbed weights, and the weights that win more survive across generations. We will also build a greedy heuristic agent as a stronger baseline for evaluation.

Phase 6 (Evaluation): We will run ablation studies comparing belief-guided vs uniform sampling, RL-tuned vs hand-tuned weights, and varying search depths and sample counts. The RL-tuned agent will be benchmarked against the hand-tuned agent, greedy baseline, and random agent to show each component's contribution.

Playable Interface: We plan to build an interactive gameplay mode supporting human vs AI, human vs human, and AI vs AI with board visualization, serving as both a demo and a tool for qualitative behavior analysis.

References

- [1] J. Blüml, J. Czech, and K. Kersting, "AlphaZero**: AlphaZero like baselines for imperfect information games are surprisingly strong," *Front. Artif. Intell.*, vol. 6, 2023. doi: 10.3389/frai.2023.1014561.
- [2] J. Yao, Z. Zhang, L. Xia, J. Yang, and Q. Zhao, "Solving Imperfect Information Poker Games Using Monte Carlo Search and POMDP Models," in *Proc. 2020 IEEE 9th Data Driven Control and Learning Systems Conf. (DDCLS)*, Liuzhou, China, 2020, pp. 1060–1065. doi: 10.1109/DDCLS49620.2020.9275053.